

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Robotics) SEMESTER - 4 Summer 2025 (Regular)

Course :Bachelor of Technology (Robotics) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMES404 - STRENGTH OF MATERIALS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1. Following are the basic types of stress except (a) Tensile stress (b) Compressive stress (c) Shear stress (d) Volumetric stress
2. The relationship between Young's modulus (E), Bulk modulus (K) and Poisson's ratio (μ) is given by (a) $E=9CK/(C+3K)$ (b) $E=9CK/(2C+3K)$ (c) $E=9CK/(3C+K)$ (d) $E=9CK/(C-3K)$
3. An axially loaded bar is subjected to a normal stress of 173 MPa. The shear stress in the bar is (a) 75 MPa (b) 86.5 MPa (c) 100 MPa (d) 122.3 MPa
4. The Point of contra flexure occurs in case of (a) Cantilever beams (b) Simply supported beams (c) Over hanging beams (d) All types of beams
5. For any part of a beam subjected to uniformly distributed load, bending moment diagram is (a) Horizontal straight line (b) Vertical straight line (c) Line inclined to x-axis (d) Parabola
6. The ratio of maximum shear stress to average shear stress is $4/3$ in _____ (a) circular cross-section (b) rectangular cross-section (c) square cross-section (d) all of the above
7. The product of the tangential force acting on the shaft and radius of shaft known as (a) Torsional rigidity (b) Flexural rigidity (c) Bending moment (d) Twisting moment
8. Deflection is found out by moment area method by using (a) First moment of area. (b) Second moment of area. (c) Third moment of area. (d) None.
9. In an off centre point loaded fixed beam the fixed bending moment diagram is a (a) square (b) rectangle (c) triangle (d) trapezium
10. A solid shaft of same cross sectional area and of same material as that of a hollow shaft can resist (a) Less torque. (b) More torque. (c). Equal torque. (d) Unequal torque.
11. Which of the following theorem can be used for deflection in fixed beams (a) Mohr's first theorem (b) Mohr's second theorem (c) Mohr's third theorem (d) Mohr's fourth theorem
12. In the theorem of three moments (a) both end sinking are considered (b) one end sinking is considered (c) sinking is neglected (d) sinking is modified by a factor

Q2. Solve the following.

- A) Define following terms: 1) Modular ratio, 2) Durability, 3) Reliability 4) Toughness, 5) Malleability, 6) Hardness 6
- B) A point in a strained material is subjected to stresses shown in fig. 1. Determine normal, tangential and resultant stresses across oblique plane by Mohr's circle method. Also calculate angle of obliquity. 6

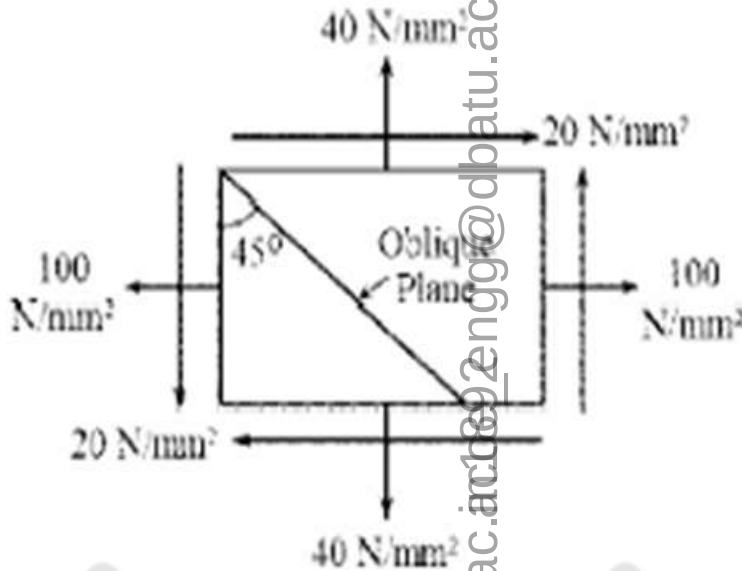


Figure 1.

Q3. Solve the following.

- A) Write the equations for Euler's crippling load for a long column with all end conditions of columns. 6
- B) A 1.5 m long column has a circular cross-section of 5cm diameter. One end of column is fixed and other end is free. Take factor of safety as 3, Calculate the safe load using i) Rankine's formula, Take yield stress as 560 N/mm^2 & $\alpha=1/1600$ for pinned ends. ii) Euler's formula, Young's modulus for CI =120 GPa. 6

Q4. Solve Any Two of the following.

- A) Derive an expression for Flexural equation along with assumptions. 6
- B) A rectangular beam 300 mm wide and 200 mm deep is simply supported over a span of 8 m. What u.d.l. per meter the beam may carry, if the bending stress is not to exceed 120 N/mm^2 6
- C) A Hollow shaft with diameter ratio of 3/8 is required to transmit 500 kW at 100 rpm, the maximum torque being 20% greater than mean. The maximum shear stress is not to exceed 60 N/mm^2 and the twist in the length of 3 m is not to exceed 1.4° . Calculate the minimum diameter required for the shaft. Take $G = 84 \text{ N/mm}^2$. 6

Q5. Solve Any Two of the following.

- A) A cantilever beam of span 5m carries a point load of 20 kN at free end and uniformly distributed load of 10 kN/m at distance of 4meter from the fixed end. Draw shear force diagram and bending moment diagram. 6

- B) A simply supported beam AB of span 4 m carries an uniformly distributed load as shown in figure 2 Draw S.F. and B.M. diagrams for the beam as shown in figure 2. Find magnitude of maximum B.M.

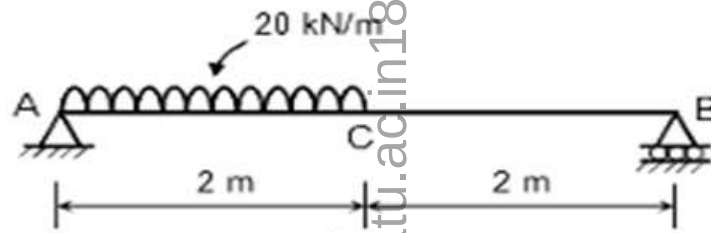


Figure 2

- C Draw shear force and bending moment diagrams for the cantilever beam as shown in figure 3.

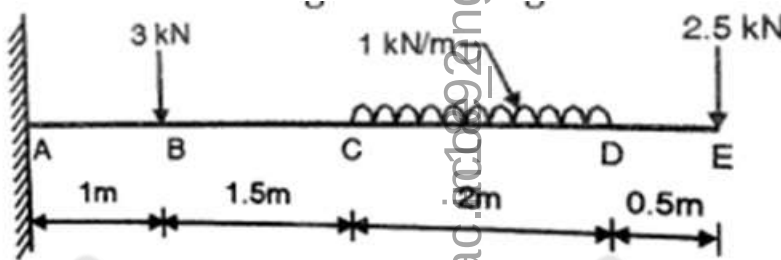


Figure 3.

Q6. Solve Any Two of the following.

- A) A simply supported beam AB of Span 5 m is carrying a point load of 30 kN at a distance 3.75 m from the left end A. Calculate the slopes at A and B and deflection under the load. Take $EI = 26 \times 10^7 \text{ N-mm}^2$.

- B) A cantilever of length 2 m carries a point load of 20 kN at the free end and another load of 20 kN at its center. If $E = 105 \text{ N/mm}^2$ and $I = 108 \text{ mm}^4$ for the cantilever then determine by moment area method, the slope and deflection of the cantilever at the free end.

- A simply supported beam 5 meter long carries 10 kN of load at the mid-point. Calculate by moment area method, the slope at the supports and deflection at the mid-span. Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $I = 1 \times 10^6 \text{ mm}^4$.

*** End ***